

Fig. 8: An actual-size, single-side PCB for storing data on I²C EEPROM using AVR microcontroller

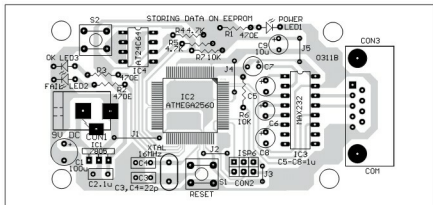


Fig. 9: Component layout for the PCB

'Erase device before programming' and 'Verify device after programming' options. Then press 'Program' to download your hex code into the ATmega2560 microcontroller.

Now you can test the code in HyperTerminal through the serial port after pressing the reset switch.

Setting HyperTerminal for testing

Start the HyperTerminal application from the desktop of your computer. Select the active COM port available on your system. Set HyperTerminal for serial communication as follows:

1. Connect the serial device (microcontroller's serial port) to the COM port of the PC.
2. Launch HyperTerminal from Start→Programs→Accessories→Communications
3. Create a new connection by specifying the name (refer Fig. 3). Set

location information as per your requirement if configuring for the first time.

4. Configure the COM port properties (refer Fig. 4) as follows:

- Bits per second: 115200
- Data bits: 8
- Parity: None
- Stop bits: 1
- Flow control: None

Once the connection is successfully established, you get a screen as shown in Fig. 5.

6. Select 'Send a text file' option from 'Transfer' menu to send the text file to the I²C EEPROM.

In this project, the communication between your PC and EEPROM starts with 'A' character. A text file can be sent to the EEPROM when it starts with 'A' character. The code can be modified for any other character.

Let's consider a 4096-byte long text file starting with 'A' followed by a

repeating EFY sequence (A EFYEFYEFYEFYEFYEFY). Send it by pressing 'Send Text' file option from 'Transfer' menu through HyperTerminal of Windows XP or lower version.

The code in the microcontroller waits for character 'A' from the PC's COM port at 115.2 kbps. As soon as it receives 'A,' it sends character 'W' to your PC as an acknowledgment. Then the rest of the characters are sent as a text file to the EEPROM through HyperTerminal. The text file is written onto the EEPROM, which sends an acknowledgement to the PC after writing is done. The text written on the EEPROM is shown in the HyperTerminal window on the screen. LED2 glows if data transfer fails, while LED3 glows to indicate that data transfer is successful. Sending a 4096-byte text file to the EEPROM and receiving the acknowledgement may take five to 10 seconds.

You can transfer the text file to the EEPROM through COM port using any serial communication software such as Terminal v1.9b, which is freely available on the Internet. Set the communication parameters as in HyperTerminal (refer Fig. 6). Press 'Connect' button followed by 'SendFile.' Now select any 4096-byte long text file that starts with 'A' from your computer and press 'Send' button to send it to EEPROM.

EEPROM verification

In order to ensure that the text is actually stored on the EEPROM, use a programmer such as TopView from Front Line Electronics. Insert the EEPROM in the ZIF socket and run the TopView programmer software. Select Device→Read, to read the data from the EEPROM. Now select EEPROM Buffer→Edit in 'Buffer' menu. You can see the data in the I²C EEPROM as shown in Fig. 7.

Construction

An actual-size, single-side PCB for storing data in the AT24C64 EEPROM using ATmega2560 AVR microcontroller circuit (Fig. 2) is shown in Fig. 8 and its component layout in Fig. 9.